

TECHCORP DUEL GAME

Final Project – Java Programming

**A TURN-BASED BUSINESS
MANAGEMENT GAME**



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Project Overview

What is TechCorp Duel Game?

- Console-based decision-making game in Java
- Player manages a tech company vs AI opponent
- 10 rounds of strategic decisions
- Goal: Earn more money & reputation than AI to WIN

Key Decisions Each Round:

1. Hire Employee (Developer / Manager / Tester)
2. Start a Project (Easy / Medium / Hard / Critical)
3. Skip turn

FROM CHAOS TO STRUCTURE

✗ Wrong Way (Beginner):

Main.java

- Hire employee
- Start project
- Calculate salary
- Handle errors
- AI logic
- Print results

*** Everything in 1 file!***

✓ My Way (Layered Architecture):

Main.java : Start here



GameEngine : Controls the game flow



Company → Project → Employee : Business logic



Exceptions : Handles errors

I separated concerns into layers so each class has ONE responsibility"

EMPLOYEE HIERARCHY

Employee (Abstract)

Developer
1.2x prod
skill×5000

Manager
0.8x prod
skill×7000

Tester
0.9x prod
skill×3000

Key OOP Concepts Used:

- ✓ Inheritance
 - Developer extends Employee
- ✓ Polymorphism
 - getProductivity() overridden
- ✓ Abstraction
 - Employee is abstract class
- ✓ Encapsulation
 - private fields + getters
- ✓ Interface
 - Comparable<Employee>A

CUSTOM EXCEPTION HIERARCHY



Features:

- Error codes (FUNDS_001, PROJ_001)
- Severity levels (LOW / MEDIUM / HIGH / CRITICAL)
- Meaningful error messages

Example:

"Insufficient funds for Hire Developer.
Required: \$250,000, Available: \$100,000"



TURN FLOW EACH ROUND

Player Turn:

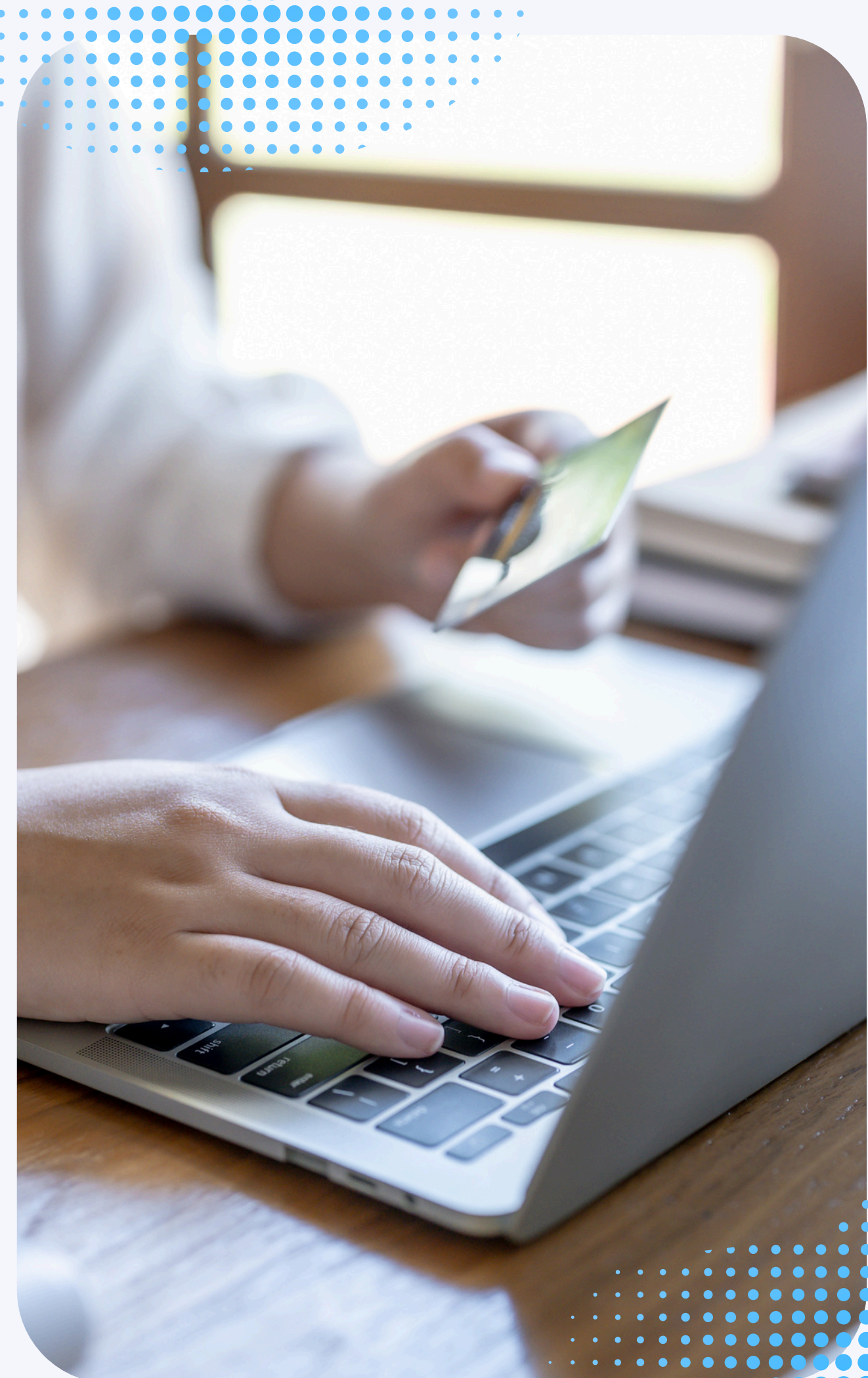
- Choose action (1/2/3)
- Validate input
- Execute action
- Deduct cash

Project Advancement:

- All IN_PROGRESS projects advance
- On deadline: success/fail by chance
- Success → add revenue + reputation

AI Turn:

- Check cash ratio
- AGGRESSIVE (>70% cash) → Hard projects
- BALANCED (>30% cash) → Medium projects
- DEFENSIVE (<30% cash) → Easy projects



STRATEGIC PROJECT CHOICES

Difficulty Budget Revenue Success
Deadline

Easy	\$5,000	\$15,000	90%	3 rounds
Medium	\$10,000	\$30,000	75%	4 rounds
Hard	\$20,000	\$60,000	60%	5 rounds
Critical	\$40,000	\$120,000	40%	6 rounds

Starting Budget: \$100,000 (Player & AI)

TECHNOLOGIES USED

- Language : Java 11
- Framework : Spring Boot 2.7.15
- Build Tool : Maven 3.9.6
- Testing : JUnit 5 (25 unit tests)
- Container : Docker
- Version Ctrl: Git & GitHub

Project Stats:

- 13 Java files
- 3 Exception classes
- 3 Employee types
- 4 Project difficulties
- 3 AI strategies

DEMO TIME !

Steps:

1. Open GitHub Codespace
2. Run: `./run.sh`
3. Play the game live!

Show:

- ✓ Hiring an employee
- ✓ Starting a project
- ✓ AI taking its turn
- ✓ Project completing
- ✓ Final score & winner

WHAT I LEARNED!

OOP Concepts:

- ✓ Inheritance & Polymorphism
- ✓ Abstract classes & Interfaces
- ✓ Encapsulation

Software Design:

- ✓ Layered Architecture
- ✓ Exception Handling
- ✓ Input Validation

Tools:

- ✓ Maven Build Automation
- ✓ Docker Containerization
- ✓ Git Version Control

The background features a light blue and white color scheme. There are three large blue circles: one in the top-left, one in the top-right, and one in the bottom-right. Each circle has a halftone dot pattern on its right side. A faint, light blue watermark of a globe with latitude and longitude lines is centered in the background.

**THANK
YOU**